

EZEQUIEL MOLINA

Economist and builder: AI adoption for education systems at global scale

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Fifteen years getting under-resourced education systems (ministries, public schools, teachers with no budget, no IT department, and no time) to successfully adopt new technology. Created global public goods used in 40+ countries (Teach: 180,000+ teachers, 3.6M students). Winner of the World Bank's inaugural Presidential Innovation Award (2025) for an AI skills-for-jobs platform, pitched personally to the Bank's President. Hands-on builder of AI evaluation systems, agentic workflows, and student-facing AI products deployed in public schools. Reach engineered with zero marketing budget: 185,000+ downloads of my AI publication series; 10,000+ weekly readers of my AI newsletter.

SYSTEMS I HAVE BUILT (HANDS-ON, SELECTED)

- *evalkit*: agentic evaluation harness for AI learning platforms. Multi-judge council reviews every test before it exists; rubric-based scoring in which every score must cite hashed evidence artifacts (re-verifiable a year later); high-stakes tests (crisis signals, minors' data) gated behind per-test human signatures. Run end-to-end against a live 3rd-grade AI math tutor used in Peruvian public schools: hallucination, cross-session state, and crisis-response behavior. Code: github.com/ezequielmolina-lang/evalkit-wip.
- *Teach + LA*: automated scoring of classroom observations with LLMs on the Teach framework I created: 92.9% agreement with certified human observers across 199 primary classrooms in Peru. Research paper under journal review.
- *El Director LIBRE / El Docente LIBRE*: AI training method for school leaders and teachers, built with Peru's Ministry of Education. 2,000+ school directors trained nationally; 27,000+ downloads since Nov 2025; RCT with 1,270 teachers (~500 treated) showing sustained AI adoption 9 months post-training; now embedded in El Salvador's \$500M education transformation (a national EdTech and AI reform) with a disbursement-linked indicator, and adapted for Brazil.
- *Eligiendo mi Camino*: AI career coach for public-school students in Lima, under evaluation across 110 public schools: 12-session program, multilingual public chat (zero-PII by design), and an AI reporting assistant for school directors. RCT evidence: pooled treatment effect of +0.13 SD on earnings-related outcomes (randomization inference $p = .005$). Cloned for Ecuador. Public replication hub: ezequielmolina-lang.github.io/emc-steal-shamelessly-hub.
- *Gallito (offline small language model)*: distilled a small language model from thousands of anonymized Claude coaching conversations so the career coach runs fully offline, as a native Windows app or in-browser on locked school Chromebooks, with nothing leaving the device. Evaluated honestly against the cloud model in my own harness: 1.55/2 vs 1.85/2 on coaching quality, hallucination rates measured and published (7.5% vs 1.2%), and one dimension where the offline model wins. Live: eligiendomicamino.org/offline.
- *APD*: autonomous generator of World Bank project designs (adapting the APE autonomous-research approach to Bank operations), scored by a blind multi-LLM judge panel with per-judge rubrics and calibrated against the Bank's official project-outcome ratings; shipped as a live design-critique tool grounded in 350+ real project documents.
- *AI Tuesday*: weekly AI-activation newsletter I write and ship myself: 10,000+ World Bank staff across all regions, 50+ editions. It is the Bank's de facto internal channel for getting non-technical staff to working AI use.
- *Adoption telemetry and playbooks*: self-built activation instrumentation for every program: an event pipeline that logs each training-module interaction across cohort tracks (thousands of rows), platform-log dormancy analysis that redesigned a national tutoring offer around actual usage instead of surveys, an auto-refreshing dashboard tracking publication reach across 213 countries, and per-cycle rollout playbooks that reduce launching each new cohort to a one-line registry change.

PROFESSIONAL EXPERIENCE

THE WORLD BANK Washington, DC · Lima · 2006 to present

Senior Economist, Regional Lead for EdTech and Teachers, Latin America (2023 to present)

- *One-to-many adoption motion, region-wide*: run the scaled funnel for AI programs across Latin America and the Caribbean: standardized products localized per country, ministry onboarding, training cohorts, embedded evaluation, and graduation into financed operations. Success measured in adoption and evidence, not fees.
- *Sales, closed*: co-lead of Peru's \$82.5M Science, Technology and Innovation project; advisor to El Salvador's \$500M transformational EdTech and AI education reform; paid advisory to Chile's Ministry of Education and Budget Office on early-

childhood financing. Rebuilt a stalled government relationship after a ministerial transition and converted initial resistance into a flagship national AI initiative, later showcased by Microsoft's CEO.

- *Ecosystem partnerships*: led the education-digitalization MoU with the Inter-American Development Bank; built a public-private coalition (Microsoft, Fundación Romero, UPC university) to scale AI teacher training nationally; working partnerships with Google, Microsoft, and Anthropic.
- *Field-to-product loop*: Presidential Innovation Award pilots now running: AI math tutors in ~70 Lima schools and AI career coaches for secondary students, solutions I designed and prototyped hands-on before handing them to implementation partners, plus AI procurement agents inside the Ministry of Education, each with evaluation designs feeding what gets scaled next.

Senior Economist / Economist, Global Lead for Teachers (2017 to 2023)

- *Product ownership at global scale*: built Teach from concept through validation to adoption in 40+ countries: 180,000+ teachers, 3.6M students, 10+ languages, taken up by UNESCO, UNICEF and NGOs, informing \$2B+ in education lending. Set direction, made the tradeoffs, shipped it, and led a 15-person global team supporting country teams in every World Bank region to improve their education systems.
- *Fundraising as GTM*: conceived Coach (teacher-coaching system, now in 15+ countries) and raised \$7M from Finland and Porticus; \$10M+ total trust funds closed through investment cases and donor pipeline management.
- *Turnaround*: took Lao PDR's \$47M national education program from 4% to 25% disbursement in one year through intensive client work; designed the results-based contract (5 disbursement-linked indicators) and the block-grant formula funding 4,500+ schools in 148 districts.

Earlier roles (2006 to 2017)

- Core team, World Development Report 2017 (chapter author); led national service-delivery surveys in Afghanistan, Tanzania and Mozambique; Gender Innovation Lab; selected into the Bank's Young Professionals Program.

EDUCATION

- Ph.D., Political Economy, Princeton University (2013). MPP, Princeton (2011). M.A. and B.A. Economics, Universidad Nacional de La Plata, Argentina (top-5 GPA award).

REACH, EVIDENCE AND RECOGNITION

- *WBG Presidential Innovation Award (2025), inaugural*: 1 of 7 winners Bank-wide, for "From Learning to Earning: Leveraging AI to Develop Skills for Jobs."
- *Distribution engineered without a marketing budget*: AI publication series with 185,000+ downloads in ~2 years: AI Revolution in Education (79,000+), AI Revolution in Higher Education (41,000+), El Director LIBRE (27,000+), Screen Time in Early Childhood (17,000+).
- *Research*: background paper to World Development Report 2026 ("Can EdTech Close Learning Gaps?", with A. Burneo, L. Dinarte-Diaz, and C. Lopez); 7 peer-reviewed articles incl. Journal of Economic Perspectives and Teaching and Teacher Education; 3ie research grant.
- 10 World Bank team awards. Work featured in The Economist, El Mercurio, Infobae, and a Microsoft CEO keynote.

TECHNICAL AND LANGUAGES

- *Daily Claude builder*: agentic workflows, eval harnesses, serverless Claude apps (vision, RAG, judge panels) designed, shipped, and maintained solo. Python, SQL, R, Stata; RCT design and impact evaluation.
- *Languages*: Spanish (native), English (fluent), Portuguese (professional).